Université Abdelmalek Essaâdi
Faculté des Sciences de Tétouan
Département Informatique

Licence IIDL / S5 :
Ingénierie Informatique et Développement Logiciel

Développement d'applications en Java

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Année universitaire 2026–2027

Élément 1

Fondamentaux et POO

- CH1 — Fondamentaux de Java
- CH2 — Programmation orientée objet
- CH3 — API Java et programmation avancée

Semaines 1 à 5

Élément 2

Bases de données et JDBC

- CH4 — Bases de données et SQL
- CH5 — JDBC
- CH6 — Transactions et sécurité

Semaines 6 à 8

Élément 3

Interfaces graphiques et architecture

- CH7 — Introduction aux IHM
- CH8 — Composants Swing
- CH9 — Architecture MVC/DAO

Semaines 9 à 12

Fil conducteur : un projet de gestion universitaire construit semaine après semaine, avec Maven, Git et l'IA de développement intégrés progressivement.

LICENCE IIDL— SEMESTRE 5

Développement d'applications en Java

Élément 2 — Bases de données et JDBC

CH4

Serveur de Base de Données

CH5

Manipulation de la Base de Données avec JDBC

CH6

Injection SQL : Attaques et Solutions

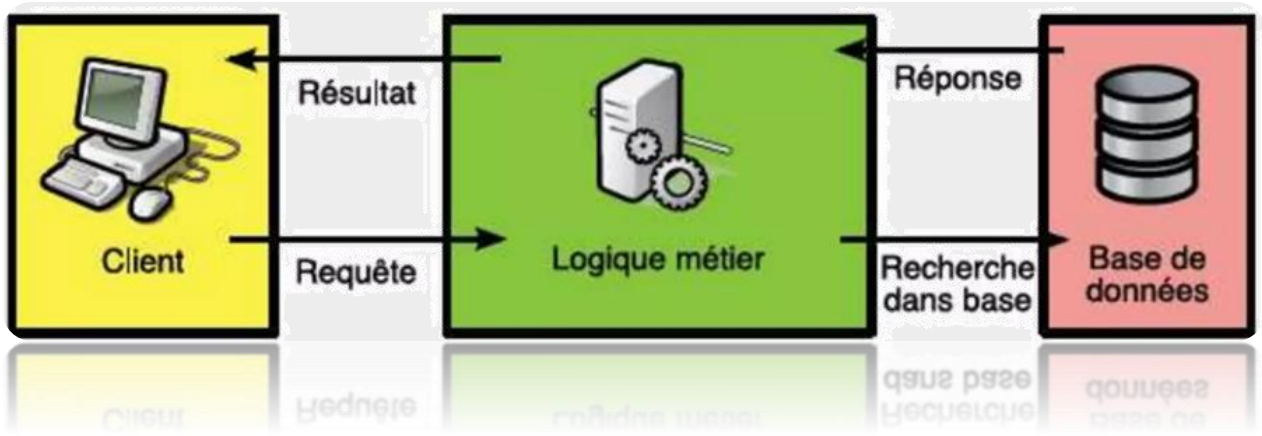
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CHAPITRE 4

Serveur de Base de Données

Serveur base de données !?

- ✓ Dans une architecture distribuée, le serveur de base de données assure la gestion et la disponibilité des données partagées entre les différents composants du système :
  - ✓ Stockage centralisé des données
  - ✓ Partage des données
  - ✓ Gestion des transactions distribuées
  - ✓ Optimisation des performances
  - ✓ Sécurité des données
  - ✓ Réplication et sauvegarde
  - ✓ Interactions avec d'autres services



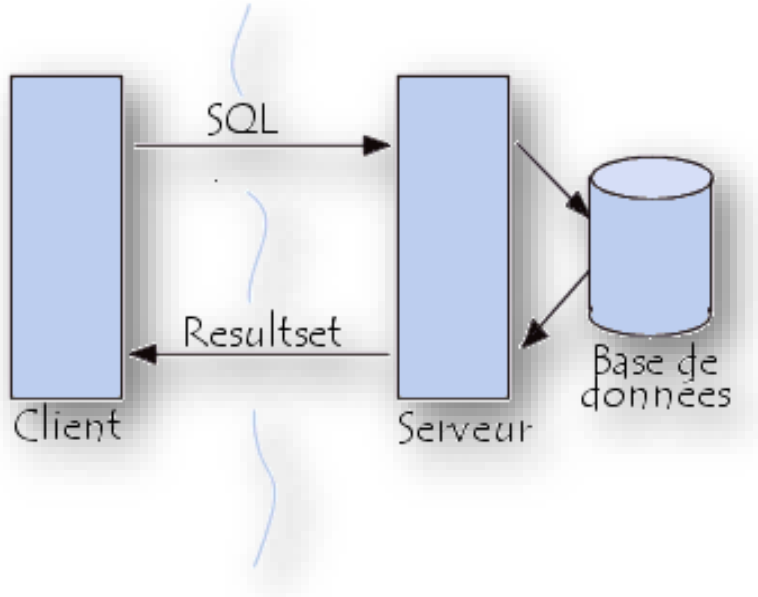
- ✓ Le serveur de base de données joue un rôle crucial au sein d'une architecture distribuée en assurant la gestion, la disponibilité, la sécurité et la performance des données partagées entre les divers composants du système. En tant que point central, il garantit la cohérence et l'intégrité des données, même dans des environnements répartis.



Serveur base de données !?

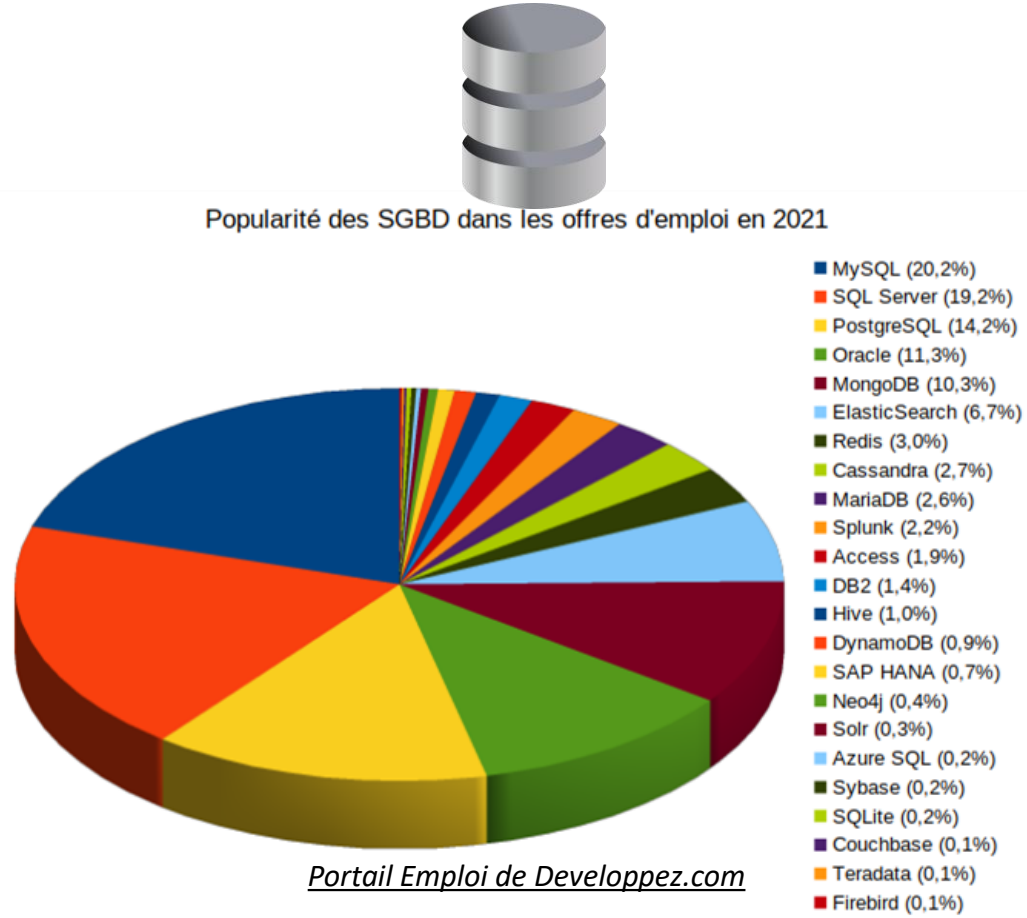
- Un serveur de base de données sert à :
  - stocker, à extraire et à gérer les données dans une base de données.
  - gérer la mise à jour des données.
  - donne un accès simultané à cette base à plusieurs serveurs d'application, Web et autres utilisateurs.
  - assure la sécurité et l'intégrité des données.
- Le logiciel de serveur de base de données offre des outils qui facilitent et accélèrent l'administration de la base, comme l'exportation de données, la configuration de l'accès de l'utilisateur et la sauvegarde des données.
- Une personne, une application ou un script de page Web peuvent interroger la base de données en recourant au langage **SQL**.





Quelle solution SGBD appropriée ?!



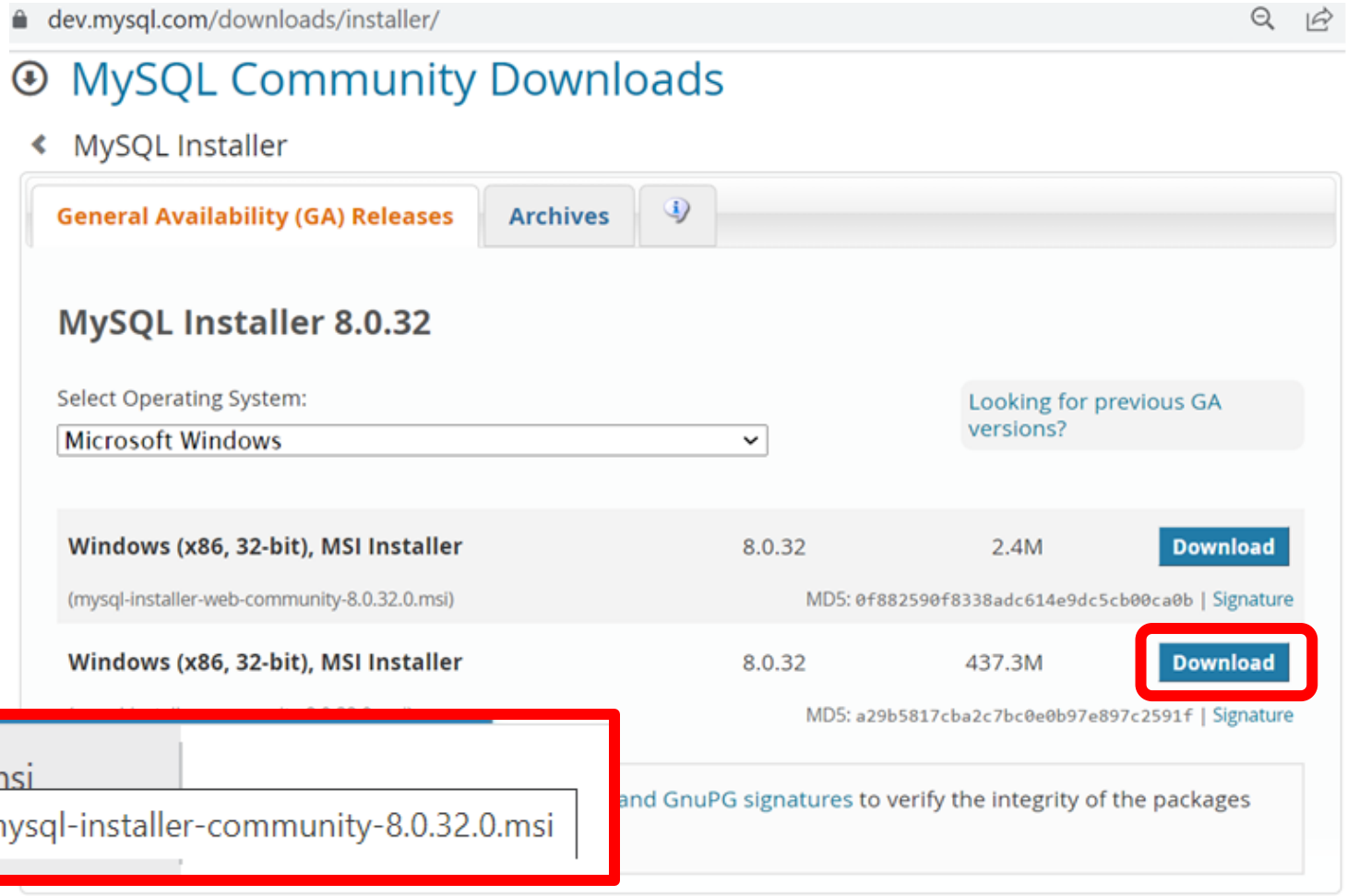


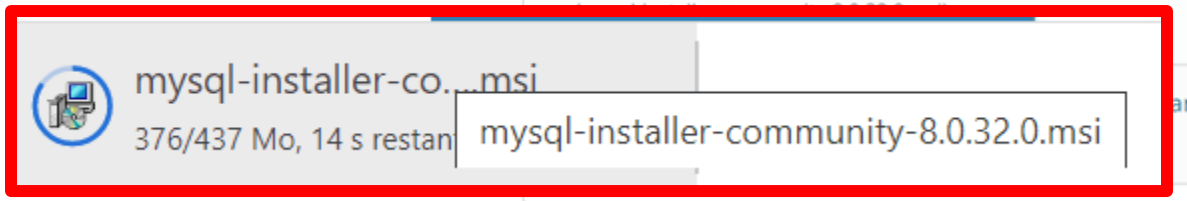
Portail Emploi de Developpez.com

- ✓ MySQL est la base de données la plus populaire en hébergement Web et développement d'application.
- ✓ MySQL est en source libre forme (gratuite), conçue pour fonctionner avec tout langage de programmation.

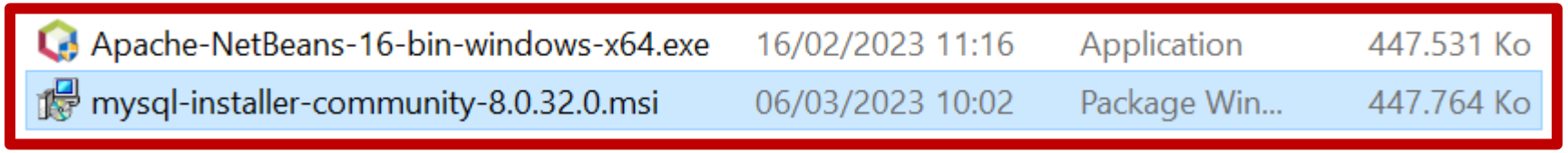
Installer un serveur DataBase MySQL ?!

- ✓ Téléchargez le paquet logiciel sur le site de MySQL : https://dev.mysql.com/downloads/installer/

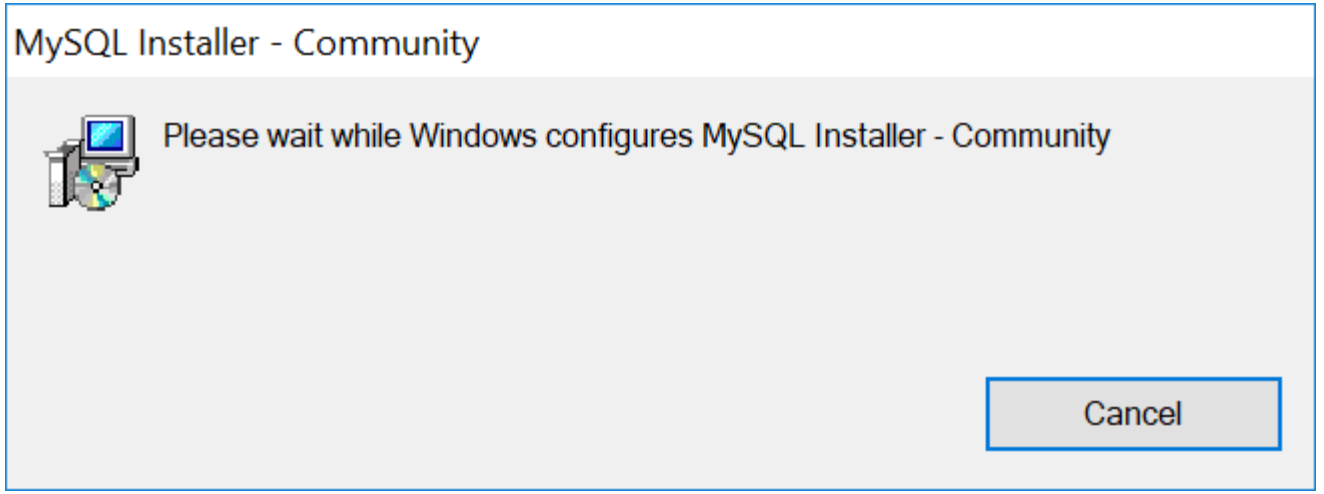




Installer un serveur DataBase MySQL ?!



- ✓ Démarrez l'installateur.



- ✓ L'installateur vous demandera de définir un mot de passe pour le super-administrateur du serveur. Veillez à bien retenir ce mot de passe, il sera utilisé plus tard.

In

✓ D

MySQL Installer

MySQL. Installer
Adding Community

- Choosing a Setup Type
- Check Requirements
- Installation
- Product Configuration
- Installation Complete

Choosing a Setup Type

Please select the Setup Type that suits your use case.

- (●) **Developer Default**
  Installs all products needed for MySQL development purposes.
- ( ) **Server only**
  Installs only the MySQL Server product.
- ( ) **Client only**
  Installs only the MySQL Client products, without a server.
- ( ) **Full**
  Installs all included MySQL products and features.
- ( ) **Custom**
  Manually select the products that should be installed on the system.

Setup Type Description

Installs the MySQL Server and the tools required for MySQL application development. This is useful if you intend to develop applications for an existing server.

This Setup Type includes:

* MySQL Server
The most popular Open Source SQL database management system.

* MySQL Shell
The new MySQL client application to manage MySQL Servers and InnoDB cluster instances.

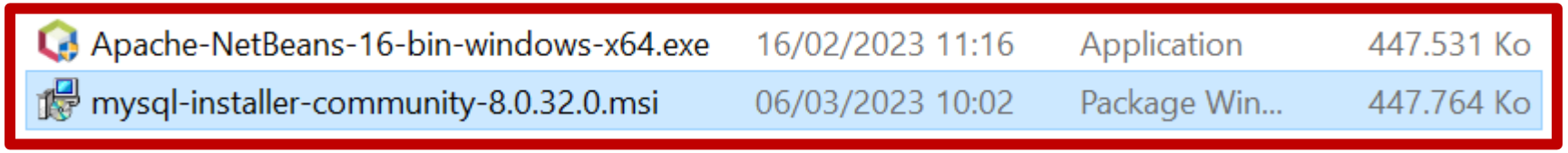
* MySQL Router

Next > Cancel

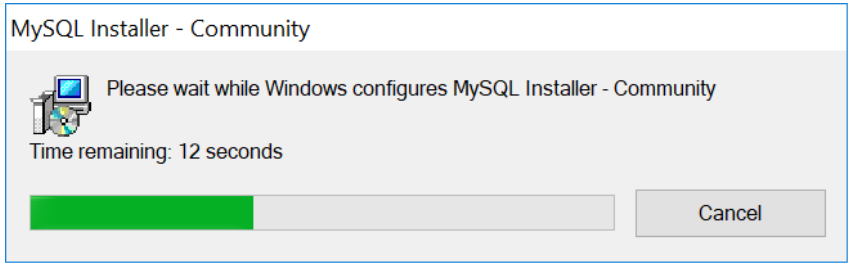
L Server. This type should
nt to deploy a MySQL
developing MySQL

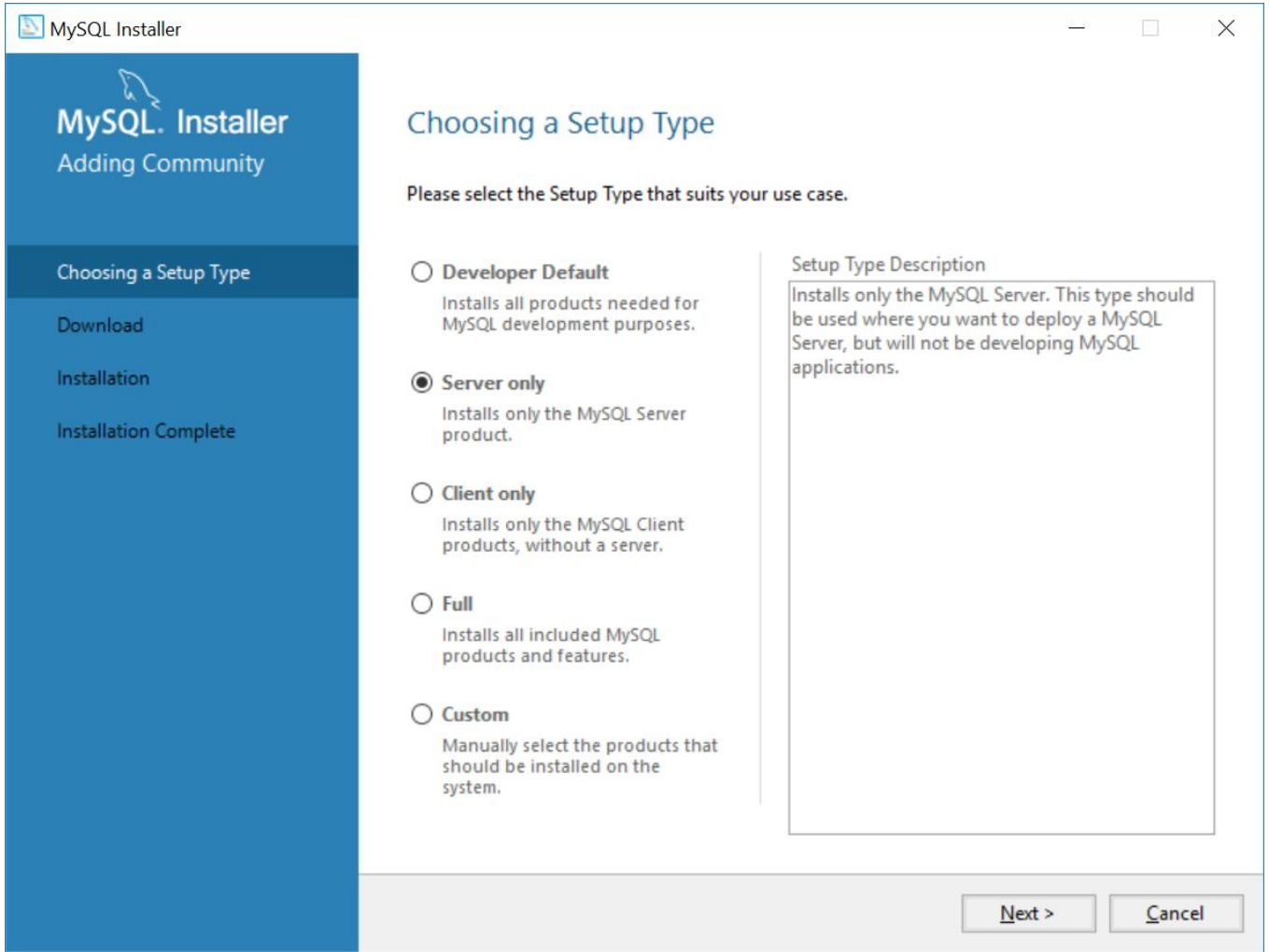
Next > Cancel

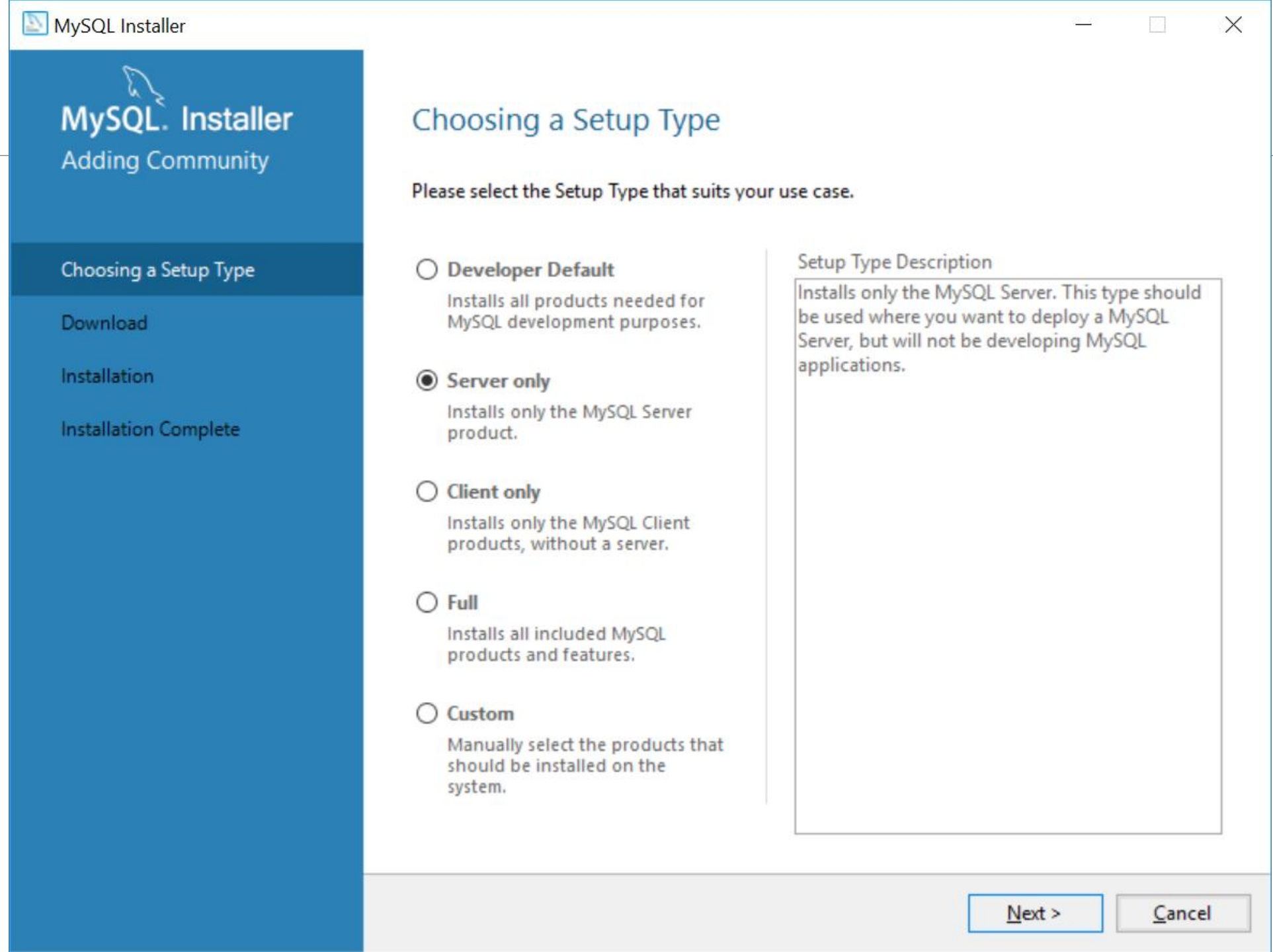
Installer un serveur DataBase MySQL ?!

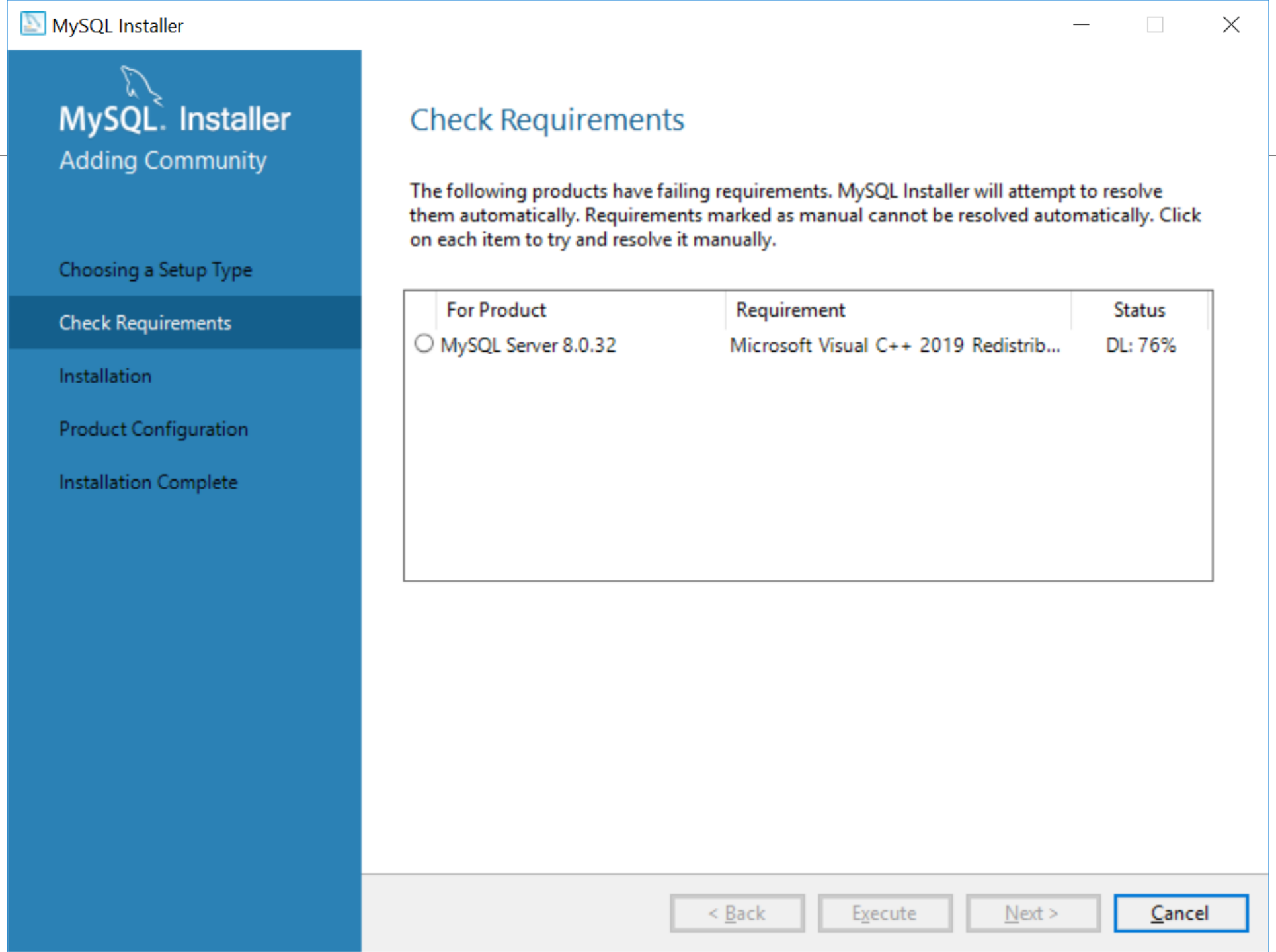


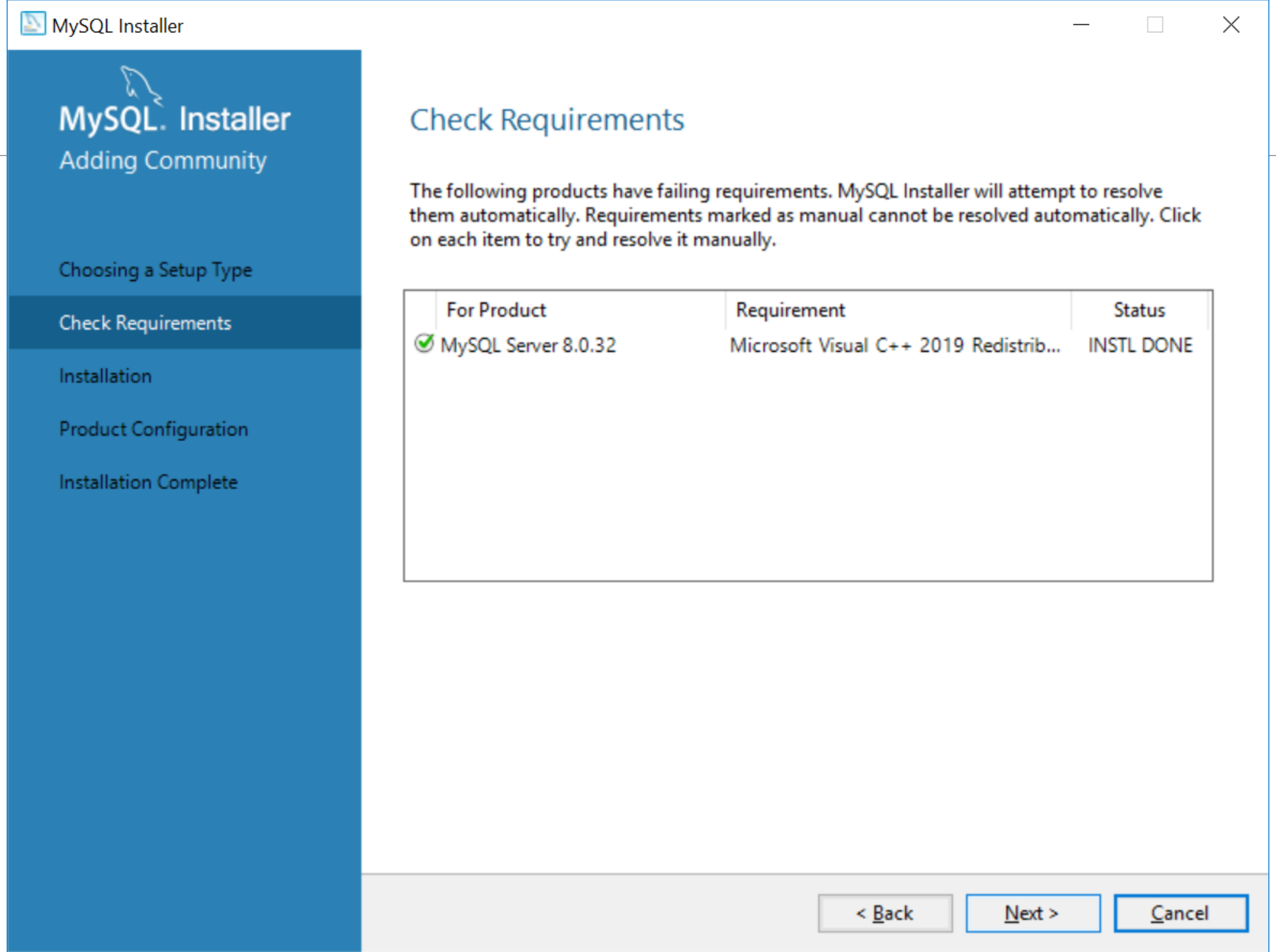
✓ Démarrez l'installateur.

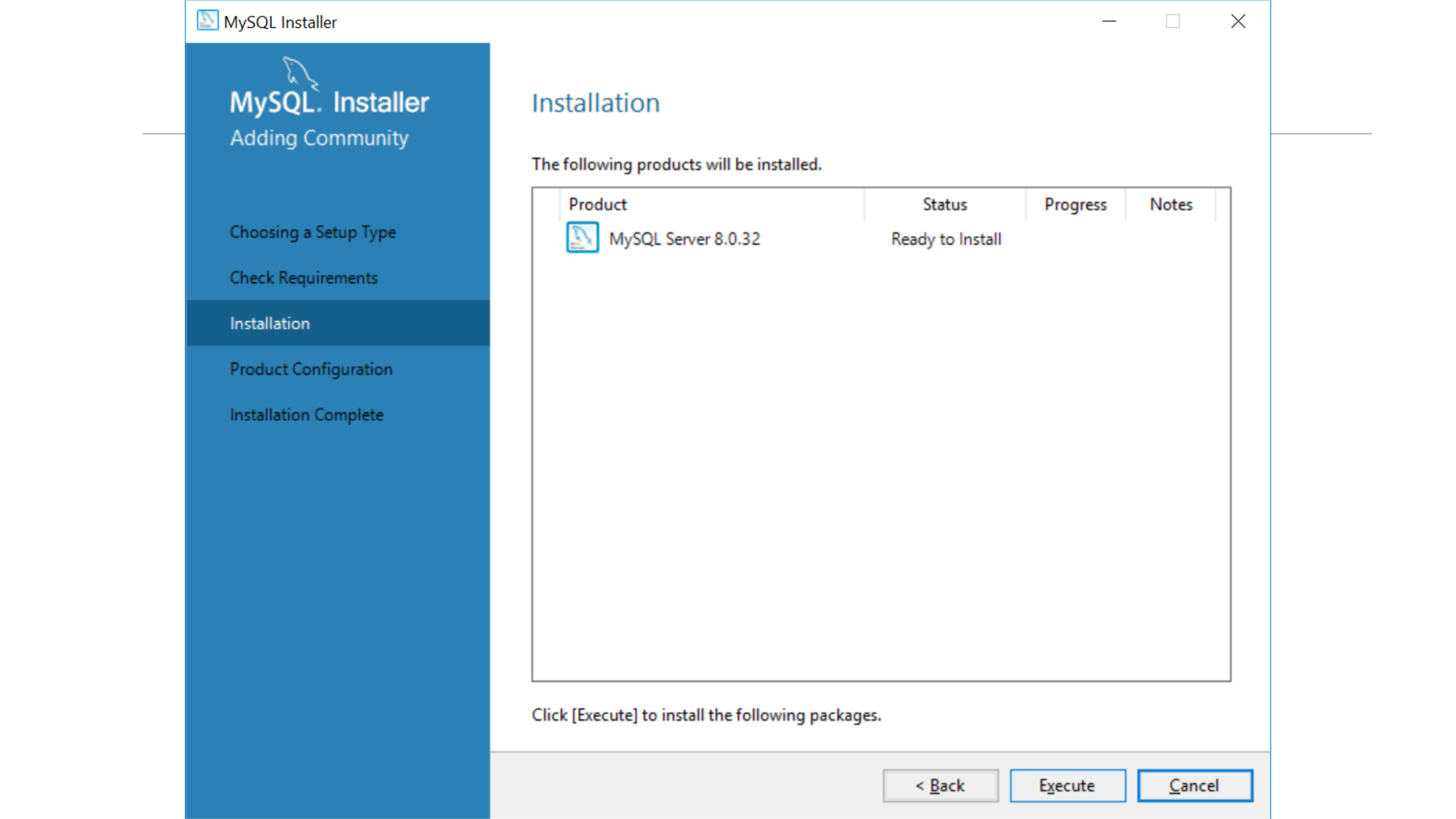


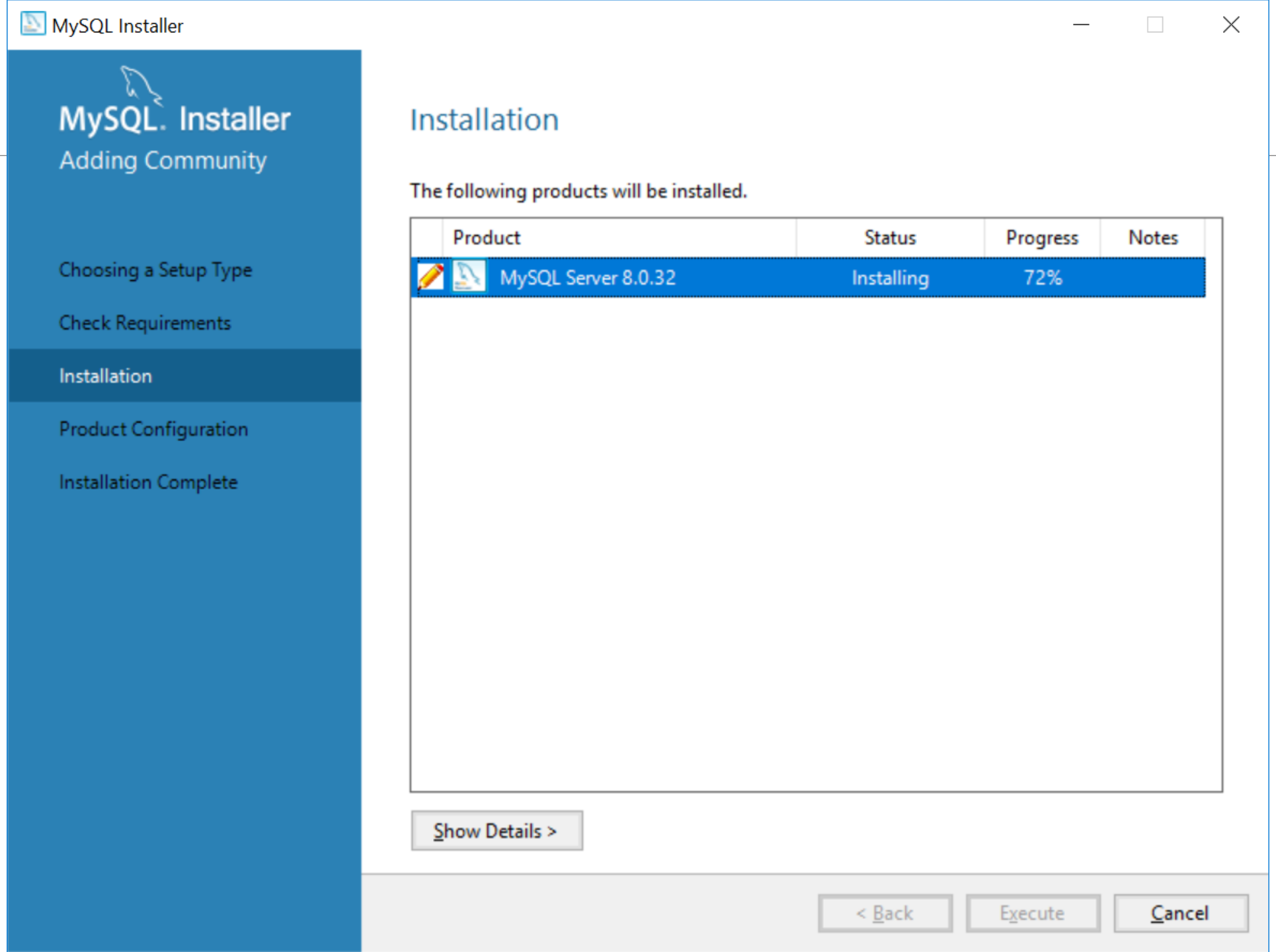


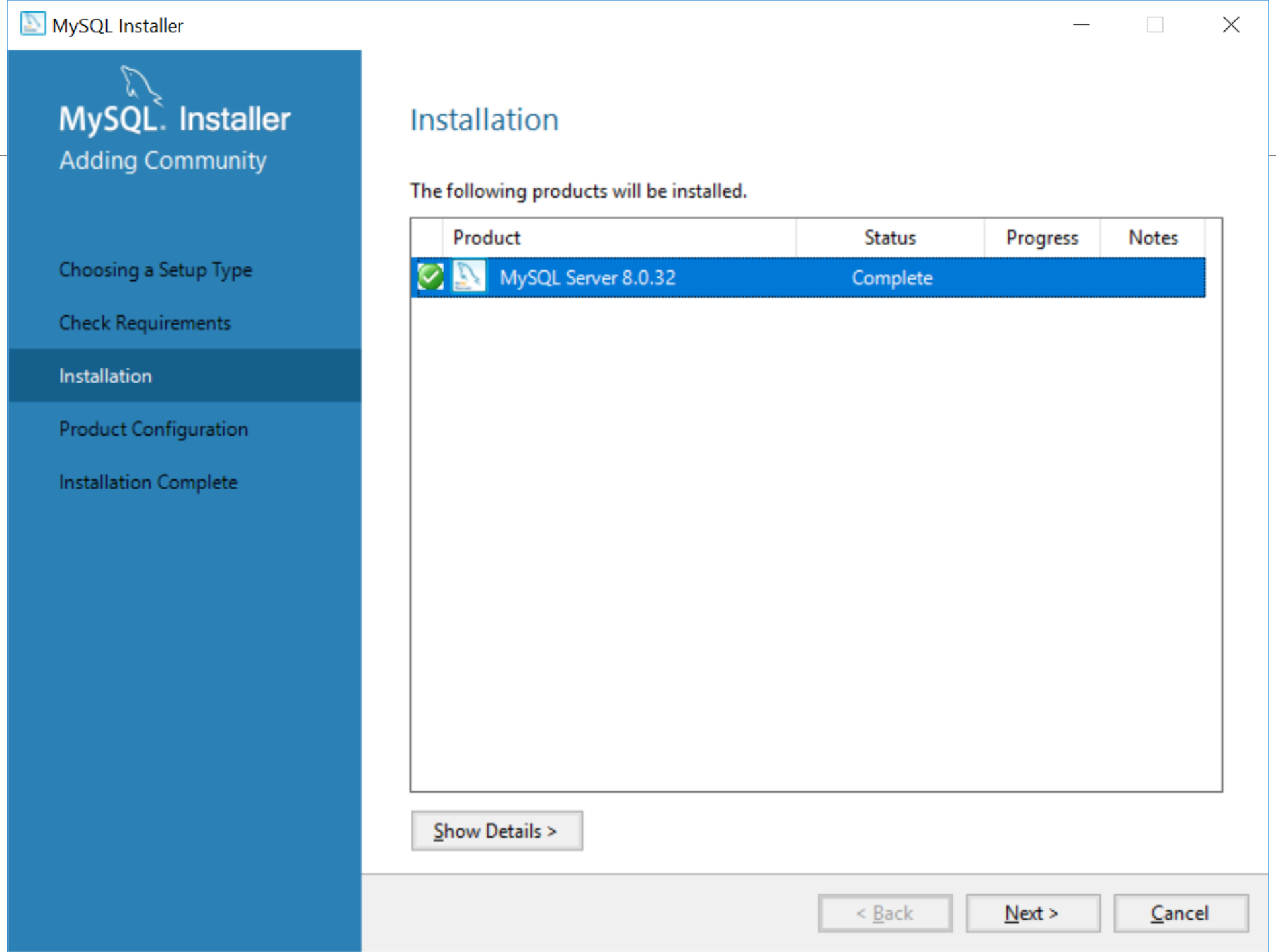

MySQL Installer
MySQL. Installer
Adding Community
Choosing a Setup Type
Download
Installation
Installation Complete
Choosing a Setup Type
Please select the Setup Type that suits your use case.
Developer Default
Installs all products needed for MySQL development purposes.
Server only
Installs only the MySQL Server product.
Client only
Installs only the MySQL Client products, without a server.
Full
Installs all included MySQL products and features.
Custom
Manually select the products that should be installed on the system.
Setup Type Description
Installs only the MySQL Server. This type should be used where you want to deploy a MySQL Server, but will not be developing MySQL applications.
Next >
Cancel

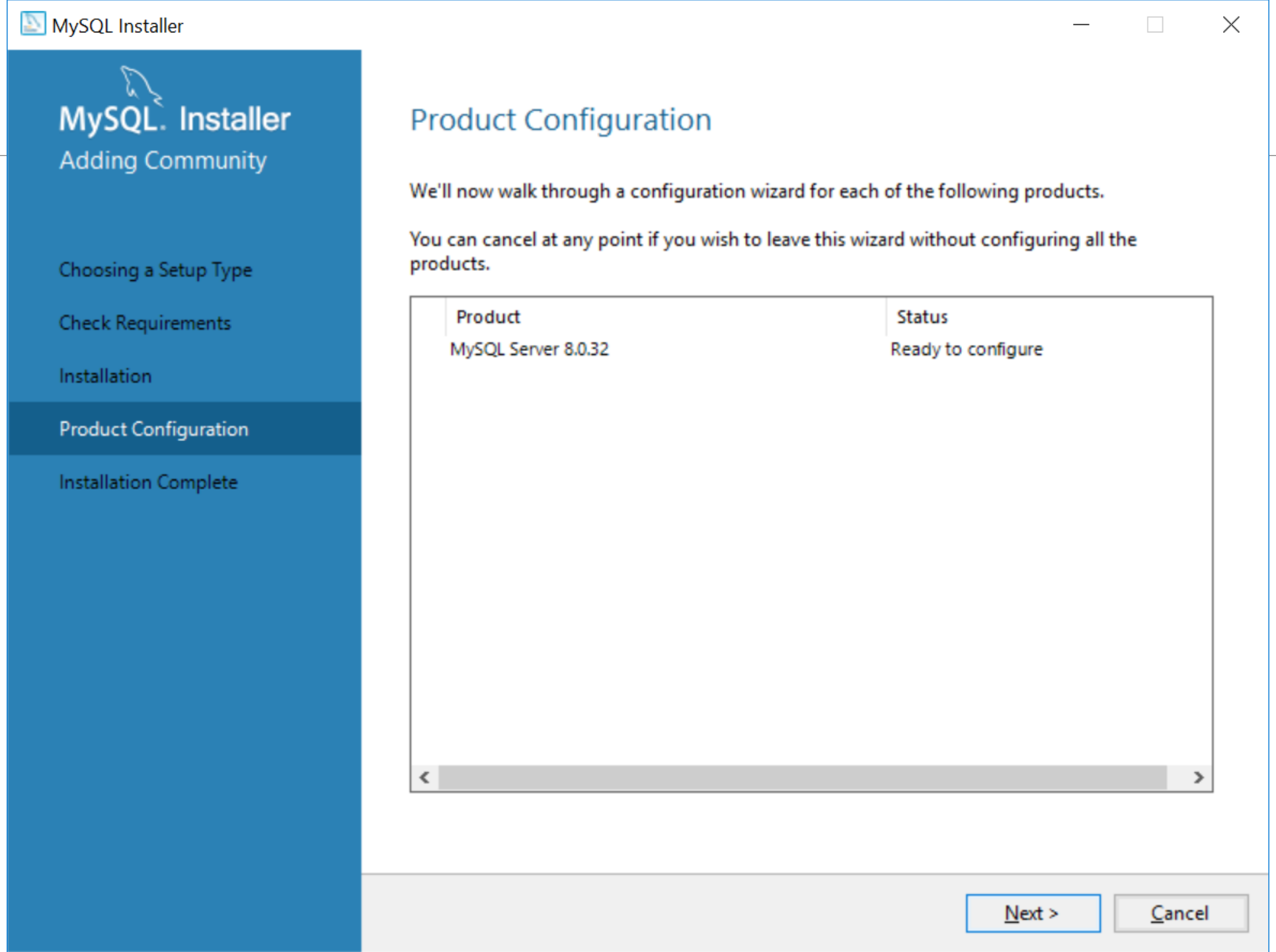

MySQL Installer
MySQL. Installer
Adding Community
Choosing a Setup Type
Check Requirements
Installation
Product Configuration
Installation Complete
Check Requirements
The following products have failing requirements. MySQL Installer will attempt to resolve them automatically. Requirements marked as manual cannot be resolved automatically. Click on each item to try and resolve it manually.
For Product
Requirement
Status
MySQL Server 8.0.32
Microsoft Visual C++ 2019 Redistrib...
DL: 76%
< Back
Execute
Next >
Cancel

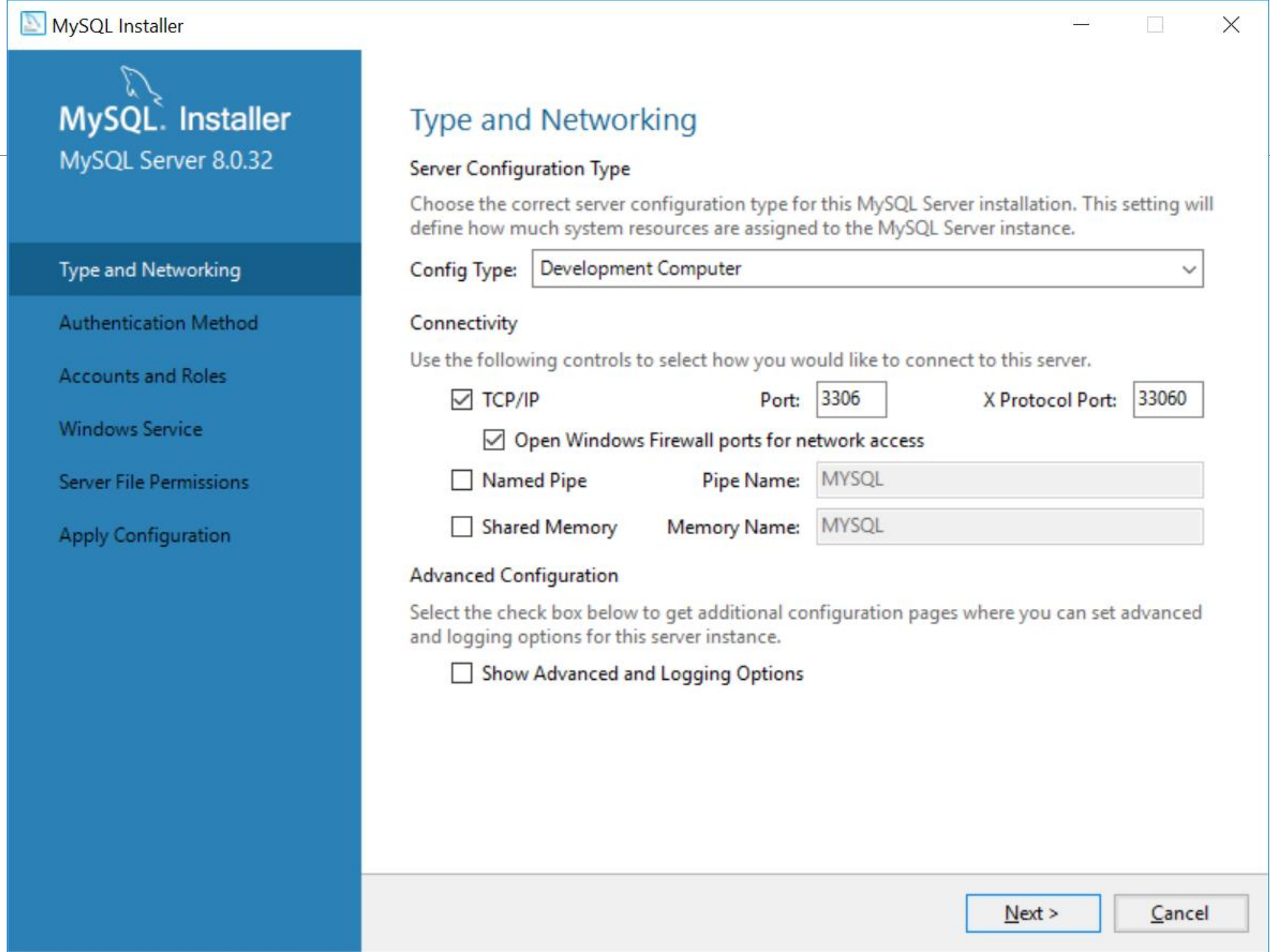

MySQL Installer
MySQL. Installer
Adding Community
Choosing a Setup Type
Check Requirements
Installation
Product Configuration
Installation Complete
Check Requirements
The following products have failing requirements. MySQL Installer will attempt to resolve them automatically. Requirements marked as manual cannot be resolved automatically. Click on each item to try and resolve it manually.
For Product
Requirement
Status
MySQL Server 8.0.32
Microsoft Visual C++ 2019 Redistrib...
INSTL DONE
< Back
Next >
Cancel


MySQL Installer
MySQL. Installer
Adding Community
Choosing a Setup Type
Check Requirements
Installation
Product Configuration
Installation Complete
Installation
The following products will be installed.
Product
Status
Progress
Notes
MySQL Server 8.0.32
Ready to Install
Click [Execute] to install the following packages.
< Back
Execute
Cancel

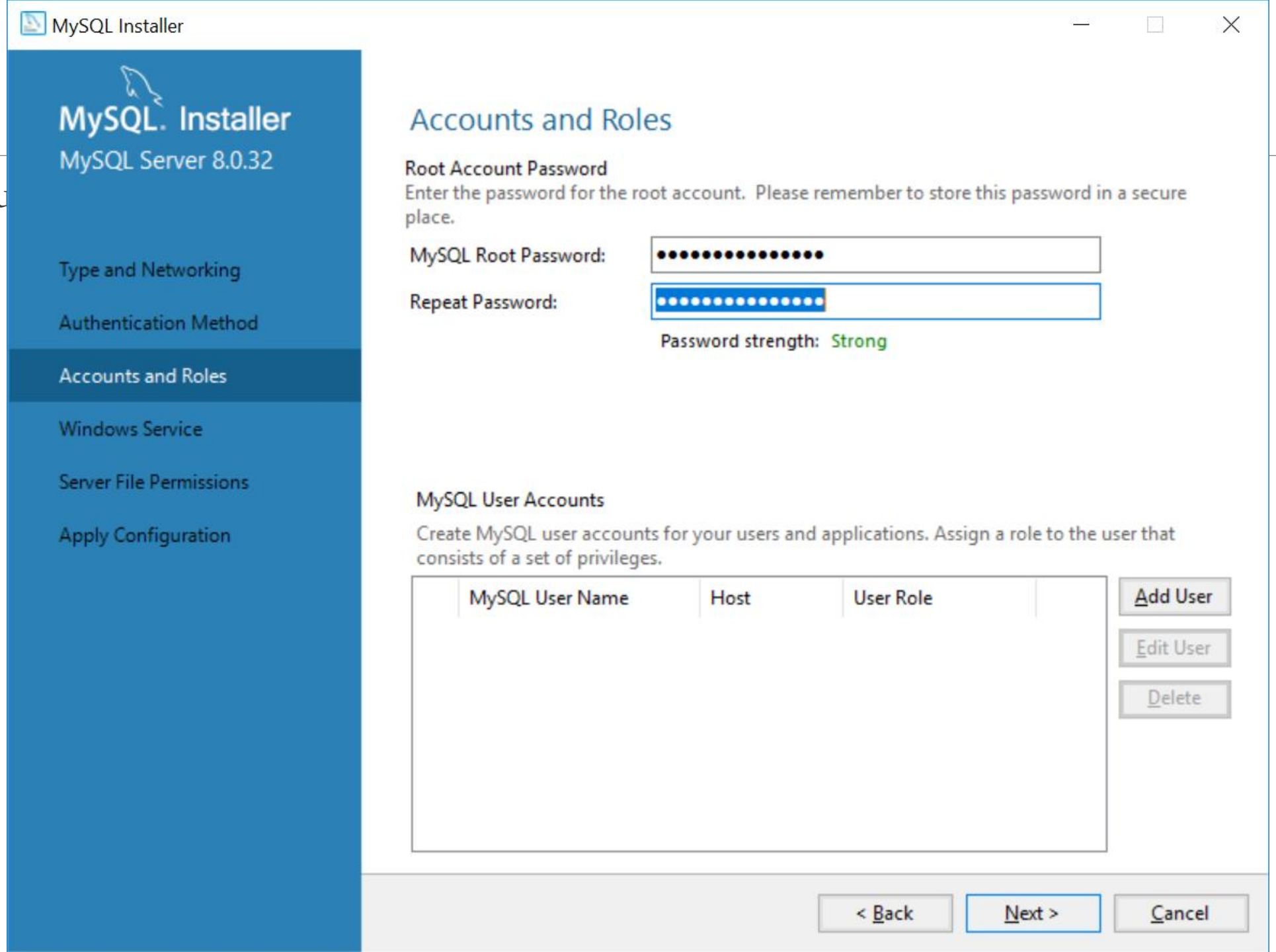

MySQL Installer
MySQL. Installer
Adding Community
Choosing a Setup Type
Check Requirements
Installation
Product Configuration
Installation Complete
Installation
The following products will be installed.
Product
Status
Progress
Notes
MySQL Server 8.0.32
Installing
72%
Show Details >
< Back
Execute
Cancel

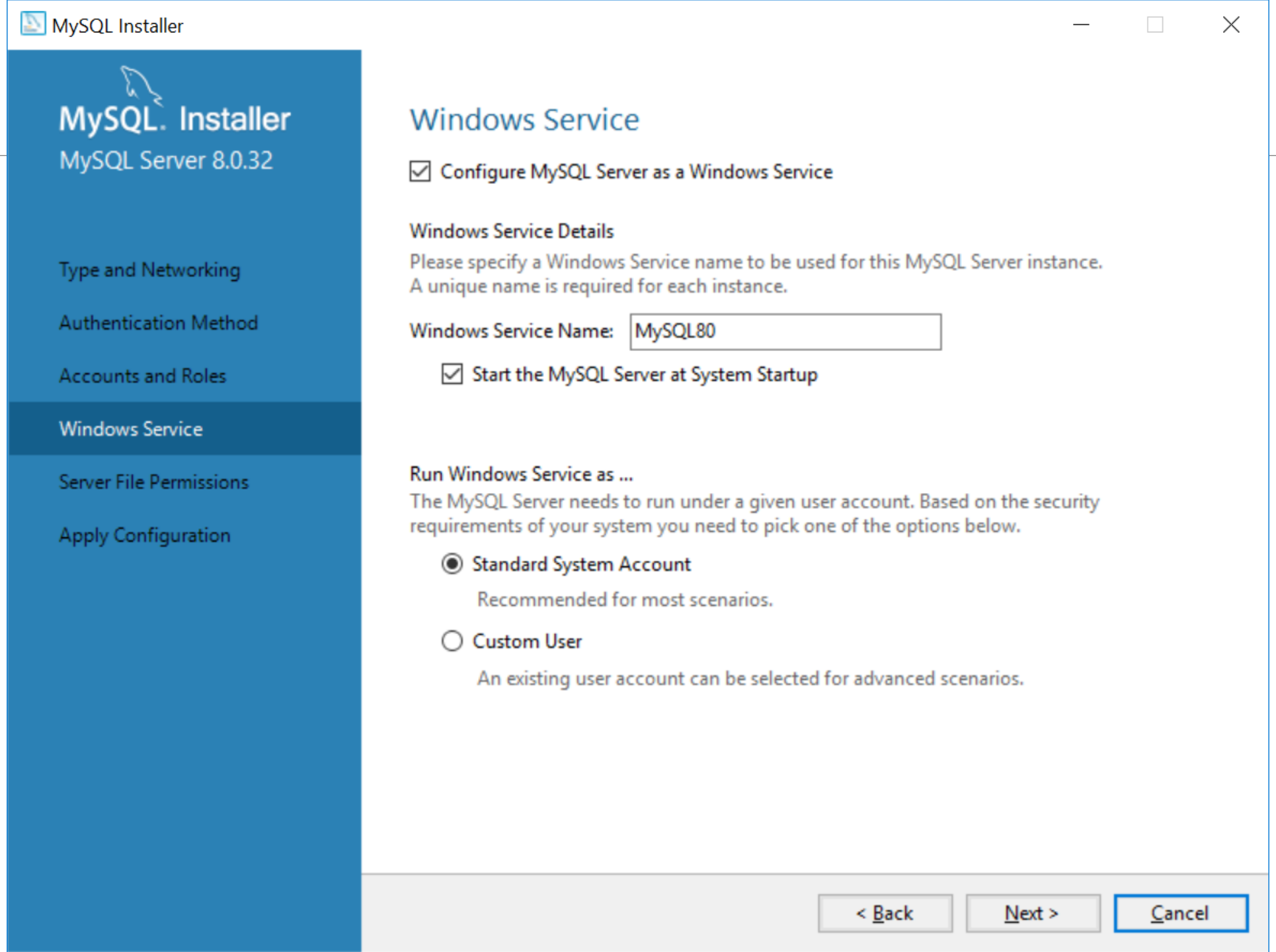

MySQL Installer
MySQL. Installer
Adding Community
Choosing a Setup Type
Check Requirements
Installation
Product Configuration
Installation Complete
Installation
The following products will be installed.
Product
Status
Progress
Notes
MySQL Server 8.0.32
Complete
Show Details >
< Back
Next >
Cancel

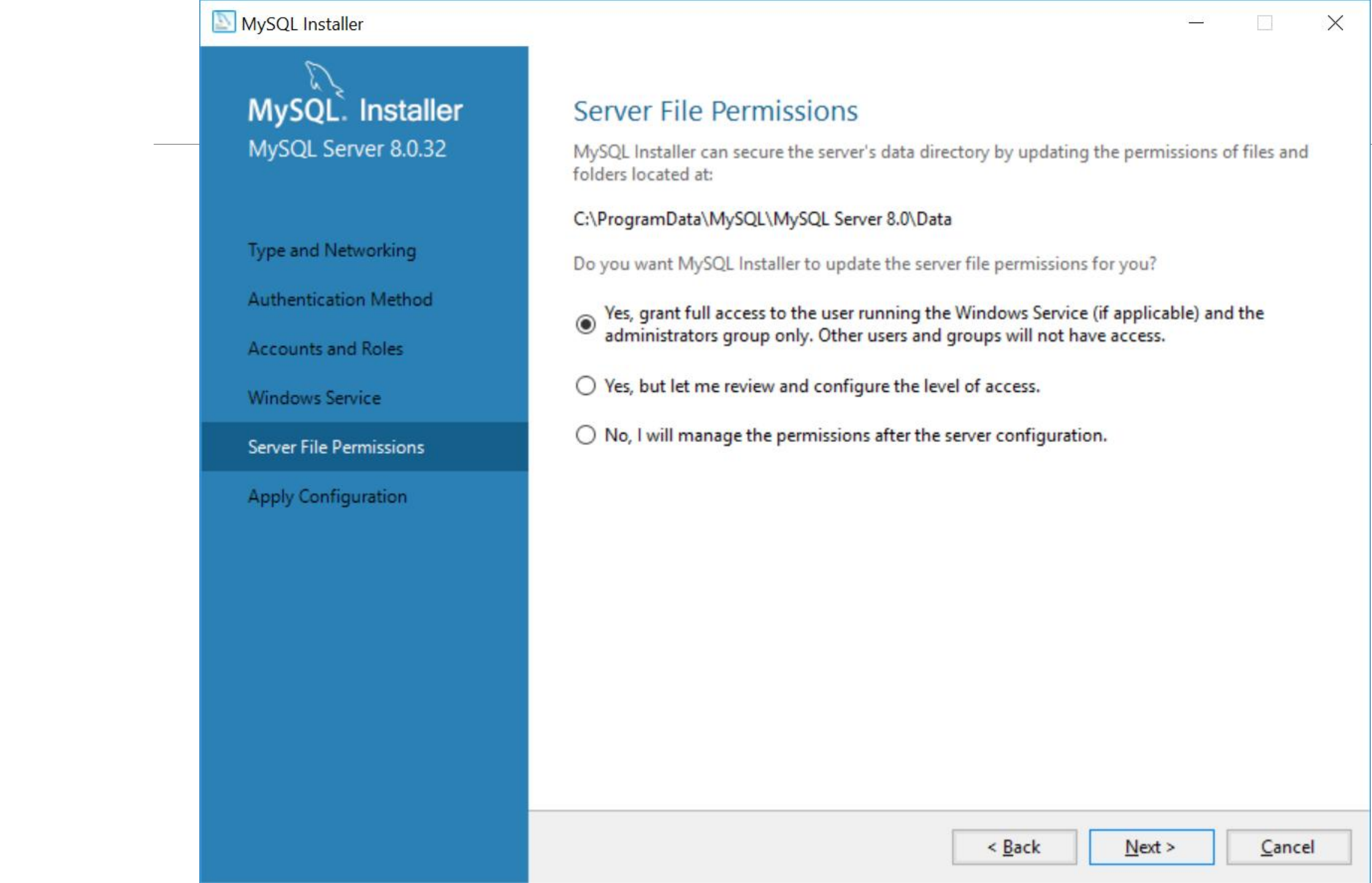

MySQL Installer
MySQL. Installer
Adding Community
Choosing a Setup Type
Check Requirements
Installation
Product Configuration
Installation Complete
Product Configuration
We'll now walk through a configuration wizard for each of the following products.
You can cancel at any point if you wish to leave this wizard without configuring all the products.
Product
Status
MySQL Server 8.0.32
Ready to configure
Next >
Cancel

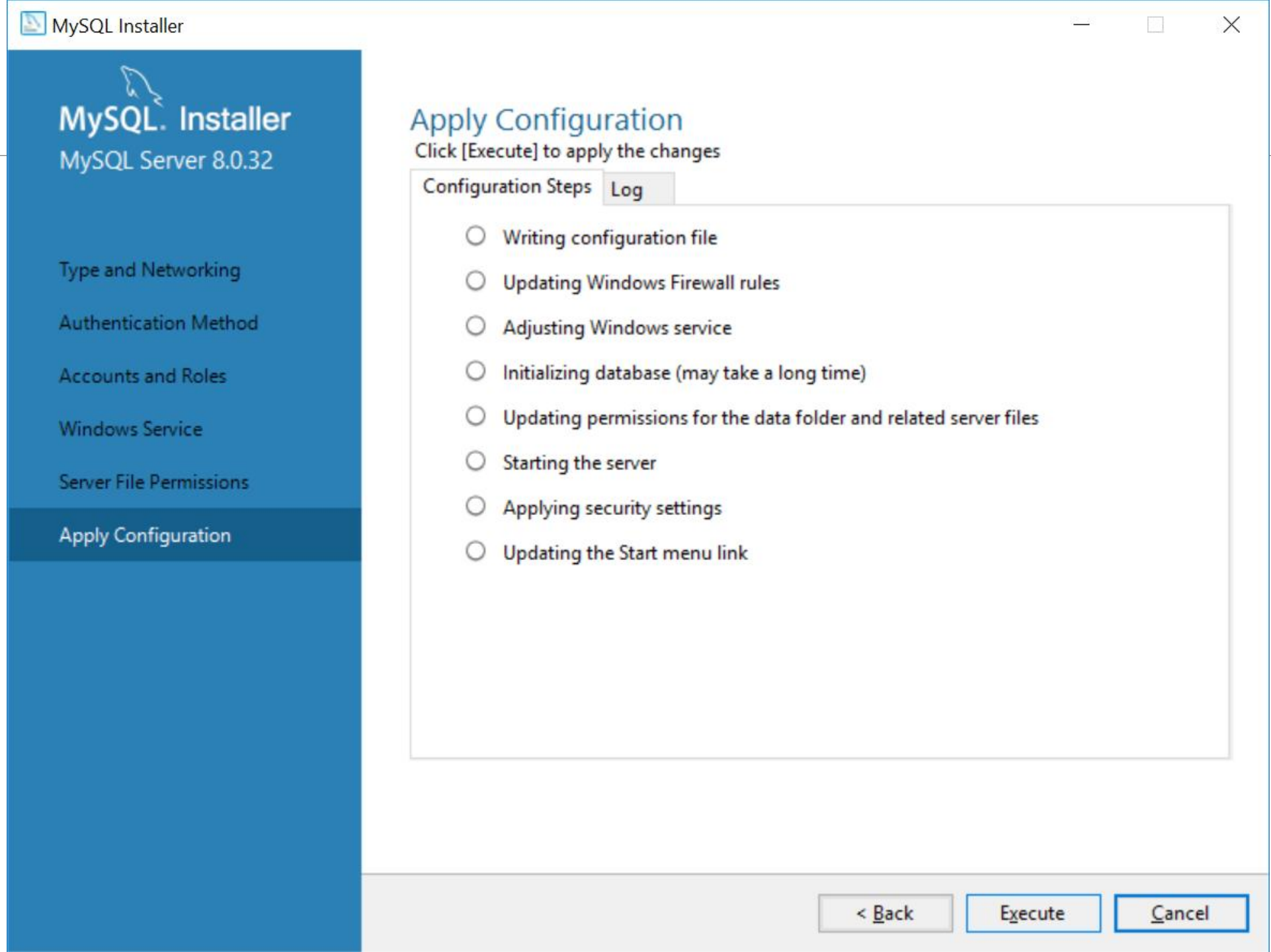

MySQL Installer
MySQL. Installer
MySQL Server 8.0.32
Type and Networking
Authentication Method
Accounts and Roles
Windows Service
Server File Permissions
Apply Configuration
Type and Networking
Server Configuration Type
Choose the correct server configuration type for this MySQL Server installation. This setting will define how much system resources are assigned to the MySQL Server instance.
Config Type: Development Computer
Connectivity
Use the following controls to select how you would like to connect to this server.
TCP/IP
Port: 3306
X Protocol Port: 33060
Open Windows Firewall ports for network access
Named Pipe
Pipe Name: MYSQL
Shared Memory
Memory Name: MYSQL
Advanced Configuration
Select the check box below to get additional configuration pages where you can set advanced and logging options for this server instance.
Show Advanced and Logging Options
Next >
Cancel

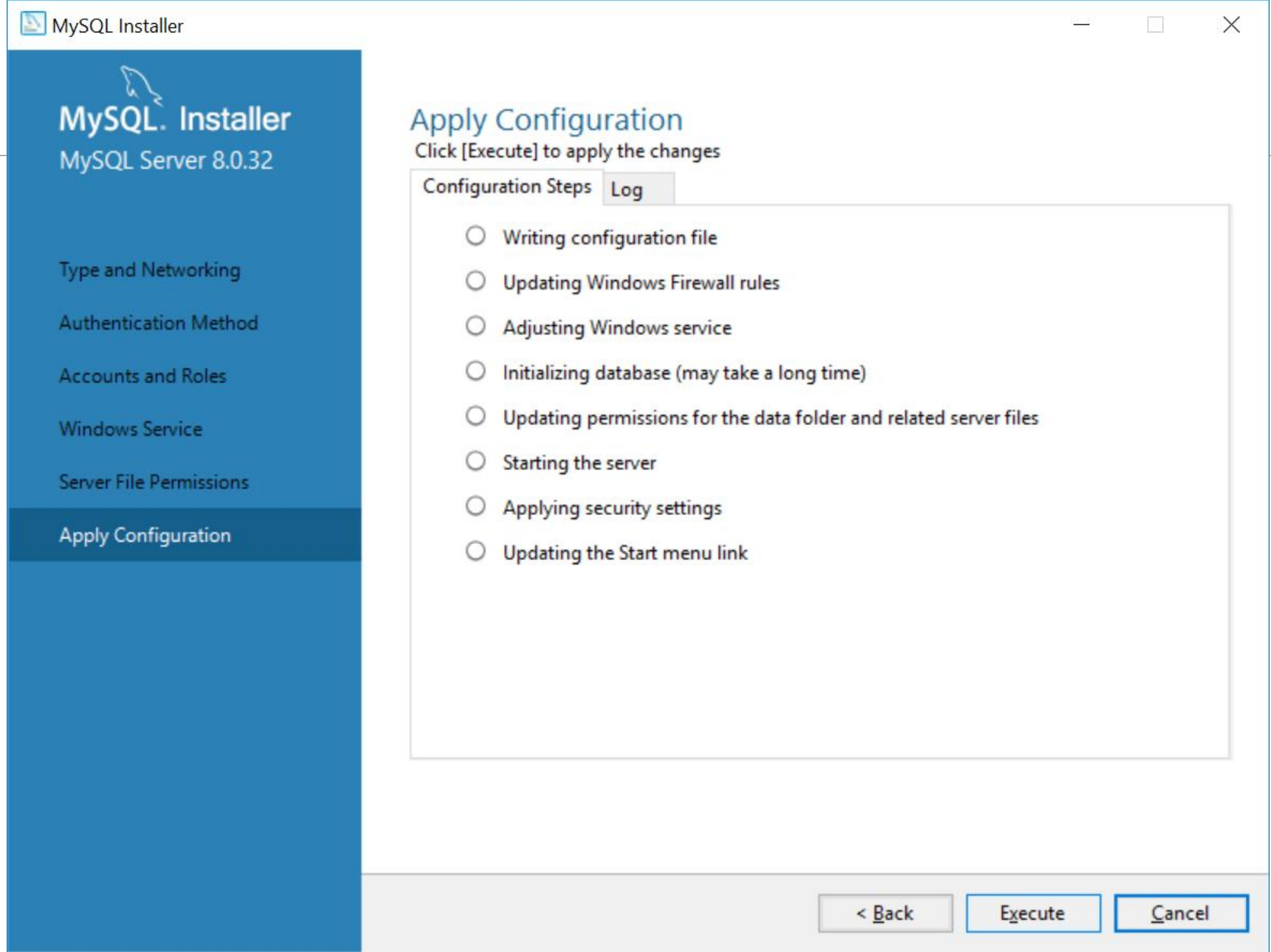
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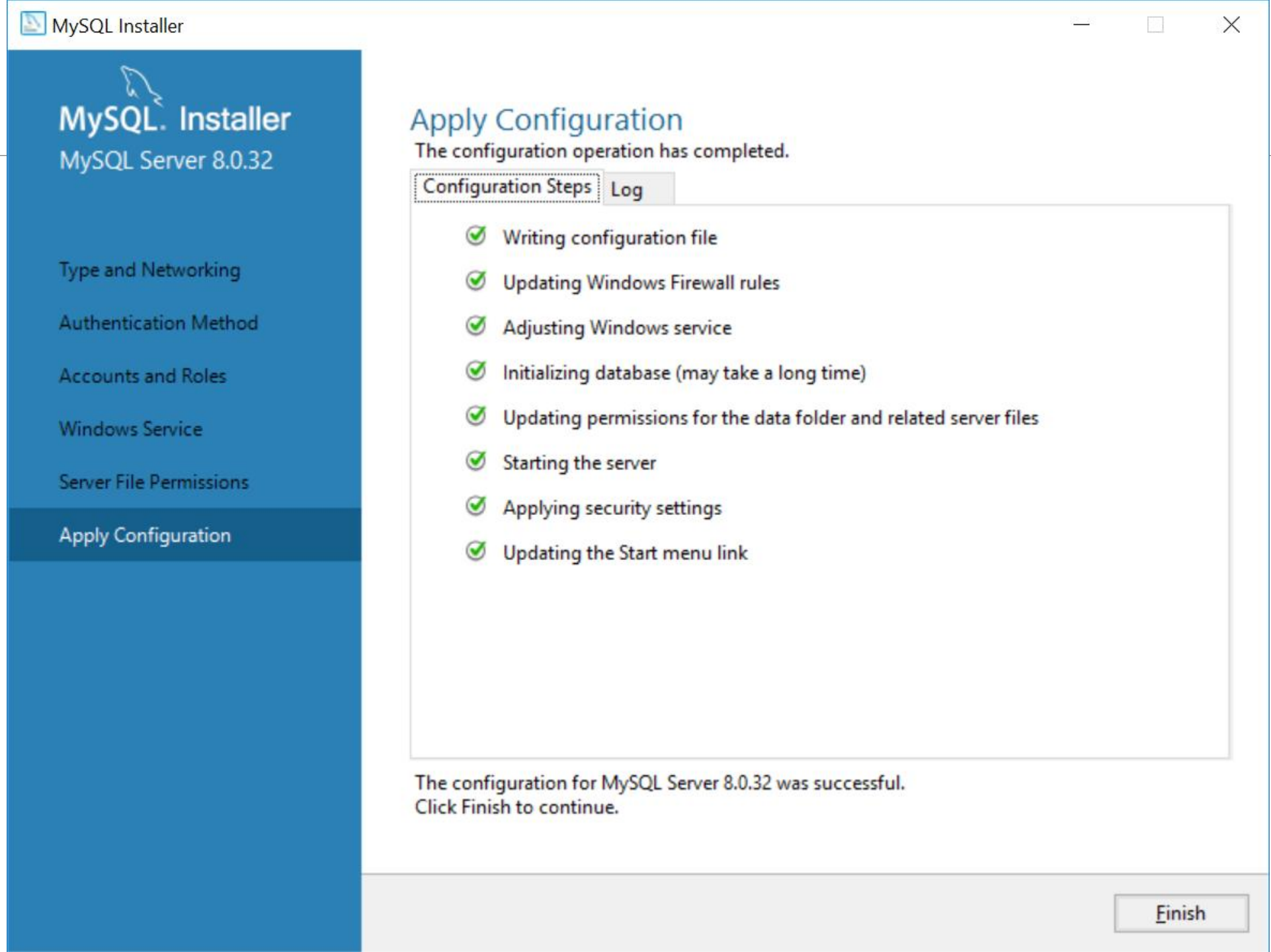

MySQL Installer
MySQL. Installer
MySQL Server 8.0.32
Type and Networking
Authentication Method
Accounts and Roles
Windows Service
Server File Permissions
Apply Configuration
Accounts and Roles
Root Account Password
Enter the password for the root account. Please remember to store this password in a secure place.
MySQL Root Password:
Repeat Password:
Password strength: Strong
MySQL User Accounts
Create MySQL user accounts for your users and applications. Assign a role to the user that consists of a set of privileges.
MySQL User Name
Host
User Role
Add User
Edit User
Delete
< Back
Next >
Cancel

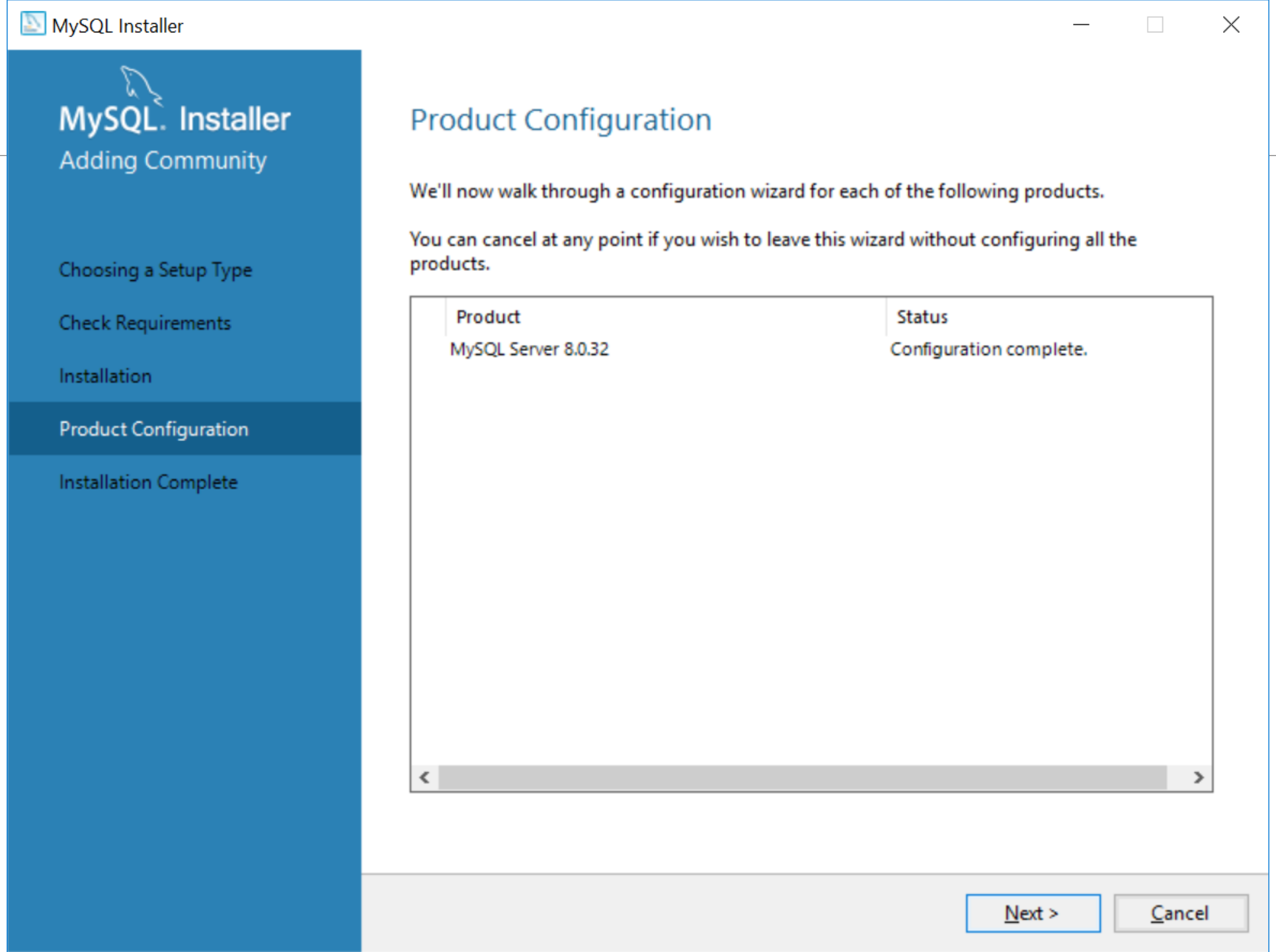

MySQL Installer
MySQL. Installer
MySQL Server 8.0.32
Type and Networking
Authentication Method
Accounts and Roles
Windows Service
Server File Permissions
Apply Configuration
Windows Service
Configure MySQL Server as a Windows Service
Windows Service Details
Please specify a Windows Service name to be used for this MySQL Server instance.
A unique name is required for each instance.
Windows Service Name:
MySQL80
Start the MySQL Server at System Startup
Run Windows Service as ...
The MySQL Server needs to run under a given user account. Based on the security requirements of your system you need to pick one of the options below.
Standard System Account
Recommended for most scenarios.
Custom User
An existing user account can be selected for advanced scenarios.
< Back
Next >
Cancel

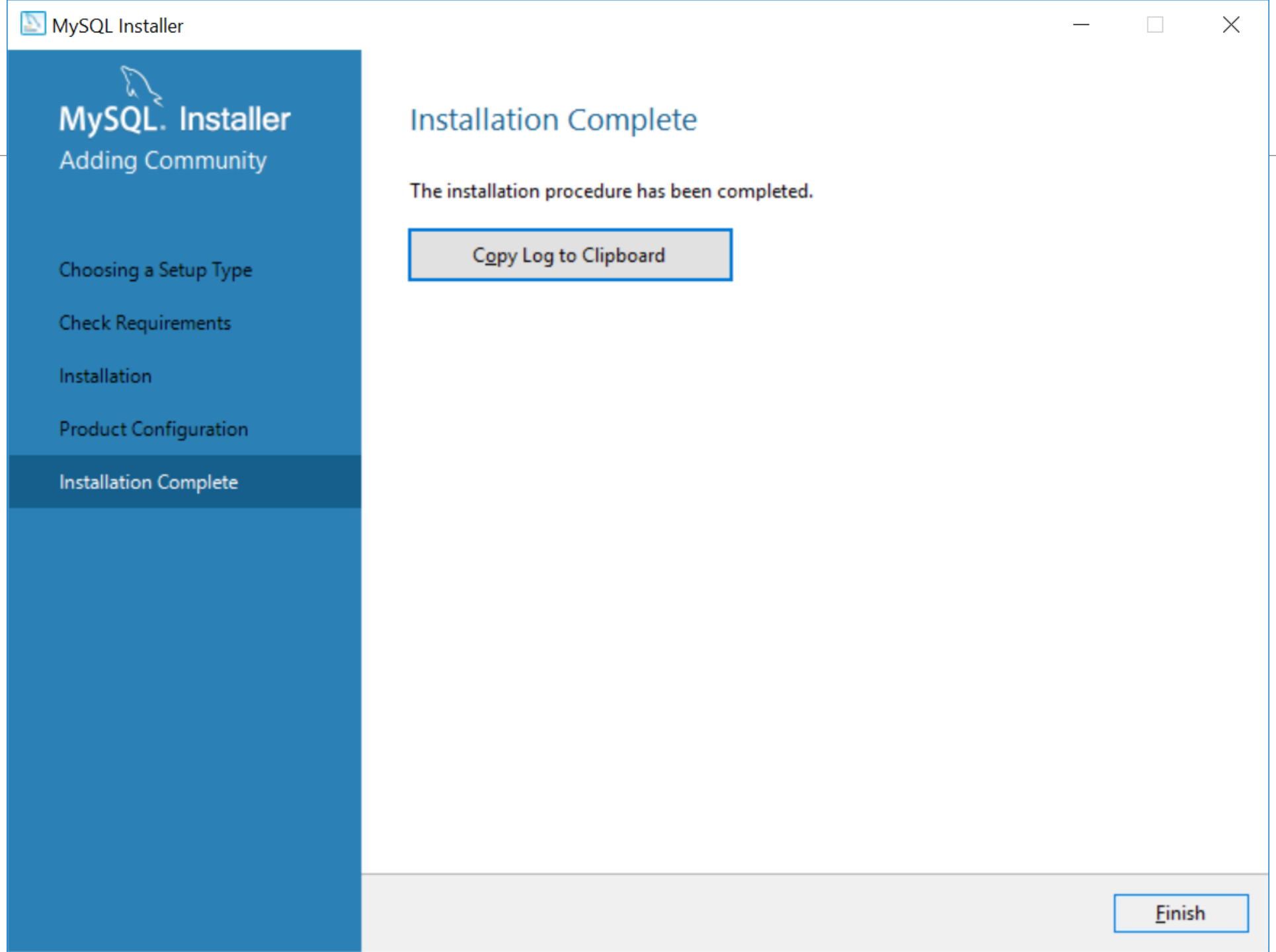

MySQL Installer
MySQL. Installer
MySQL Server 8.0.32
Type and Networking
Authentication Method
Accounts and Roles
Windows Service
Server File Permissions
Apply Configuration
Server File Permissions
MySQL Installer can secure the server's data directory by updating the permissions of files and folders located at:
C:\ProgramData\MySQL\MySQL Server 8.0\Data
Do you want MySQL Installer to update the server file permissions for you?
Yes, grant full access to the user running the Windows Service (if applicable) and the administrators group only. Other users and groups will not have access.
Yes, but let me review and configure the level of access.
No, I will manage the permissions after the server configuration.
< Back
Next >
Cancel


MySQL Installer
MySQL. Installer
MySQL Server 8.0.32
Type and Networking
Authentication Method
Accounts and Roles
Windows Service
Server File Permissions
Apply Configuration
Apply Configuration
Click [Execute] to apply the changes
Configuration Steps
Log
Writing configuration file
Updating Windows Firewall rules
Adjusting Windows service
Initializing database (may take a long time)
Updating permissions for the data folder and related server files
Starting the server
Applying security settings
Updating the Start menu link
< Back
Execute
Cancel


MySQL Installer
MySQL. Installer
MySQL Server 8.0.32
Type and Networking
Authentication Method
Accounts and Roles
Windows Service
Server File Permissions
Apply Configuration
Apply Configuration
Click [Execute] to apply the changes
Configuration Steps
Log
Writing configuration file
Updating Windows Firewall rules
Adjusting Windows service
Initializing database (may take a long time)
Updating permissions for the data folder and related server files
Starting the server
Applying security settings
Updating the Start menu link
< Back
Execute
Cancel

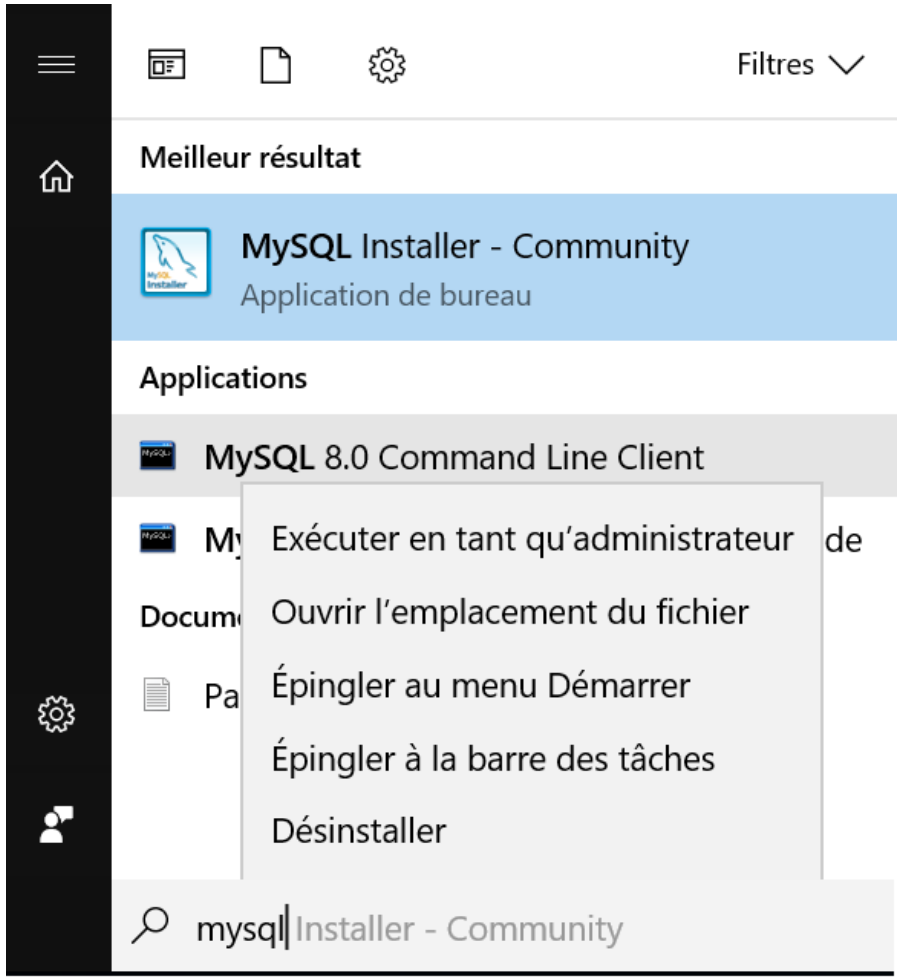

MySQL Installer
MySQL. Installer
MySQL Server 8.0.32
Type and Networking
Authentication Method
Accounts and Roles
Windows Service
Server File Permissions
Apply Configuration
Apply Configuration
The configuration operation has completed.
Configuration Steps
Log
Writing configuration file
Updating Windows Firewall rules
Adjusting Windows service
Initializing database (may take a long time)
Updating permissions for the data folder and related server files
Starting the server
Applying security settings
Updating the Start menu link
The configuration for MySQL Server 8.0.32 was successful.
Click Finish to continue.
Finish


MySQL Installer
MySQL. Installer
Adding Community
Choosing a Setup Type
Check Requirements
Installation
Product Configuration
Installation Complete
Product Configuration
We'll now walk through a configuration wizard for each of the following products.
You can cancel at any point if you wish to leave this wizard without configuring all the products.
Product
Status
MySQL Server 8.0.32
Configuration complete.
Next >
Cancel


MySQL Installer
MySQL. Installer
Adding Community
Choosing a Setup Type
Check Requirements
Installation
Product Configuration
Installation Complete
Installation Complete
The installation procedure has been completed.
Copy Log to Clipboard
Finish

Run DataBase MySQL server

Lancez MySQL



MySQL 8.0 Command Line Client

```
Your MySQL connection id is 10
Server version: 8.0.32 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| dbcafe             |
| dbemploye          |
| dbprof             |
| dbstds             |
| dbstock            |
| dbtest             |
| dbtest2            |
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
11 rows in set (0.00 sec)

mysql>
```

Créer la Base de données

- ✓ Pour créer une base, on utilise la commande :

```
CREATE DATABASE nomdelabase;
```

```
mysql> CREATE DATABASE DBTest;
Query OK, 1 row affected (0.13 sec)
```

- ✓ Supprimer Base de Données :

```
DROP DATABASE DBTest;
```

- ✓ Utiliser une base de données avec **USE** :

```
USE nomdelabase;
```

```
mysql> USE DBTest;
Database changed
```

- ✓ Créer des tables avec **CREATE TABLE :**

```
mysql> CREATE TABLE etudiant (
    -> id INTEGER NOT NULL AUTO_INCREMENT PRIMARY KEY,
    -> nom VARCHAR(100),
    -> email VARCHAR(255) NOT NULL UNIQUE,
    -> note FLOAT
    ->  );
Query OK, 0 rows affected (0.19 sec)
```

Afficher Description sur une table : **DESCRIBE**

```
mysql> describe etudiant;
+-------+--------------+------+-----+---------+----------------+
| Field | Type         | Null | Key | Default | Extra          |
+-------+--------------+------+-----+---------+----------------+
| id    | int          | NO   | PRI | NULL    | auto_increment |
| nom   | varchar(100) | YES  |     | NULL    |                |
| email | varchar(255) | NO   | UNI | NULL    |                |
| note  | float        | YES  |     | NULL    |                |
+-------+--------------+------+-----+---------+----------------+
```

Manipulation de la base de données

- ✓ Vérifier l'intégrité de votre table avec SHOW tables :

```
SHOW TABLES;
```

```
mysql> SHOW tables;
+------------------+
| Tables_in_dbtest |
+------------------+
| etudiant         |
+------------------+
```

- ✓ Supprimer définitivement une table d'une base de données :

```
DROP TABLE table_name;
```

```
mysql> DROP TABLE etudiant;
Query OK, 0 rows affected (0.06 sec)
```

- ✓ Ajouter un enregistrement dans une table de la base de données :

```
INSERT INTO table INSERT INTO table (nom_col1, nom_col2, ...)
                  VALUES ('valeur 1', 'valeur 2', ...);
```

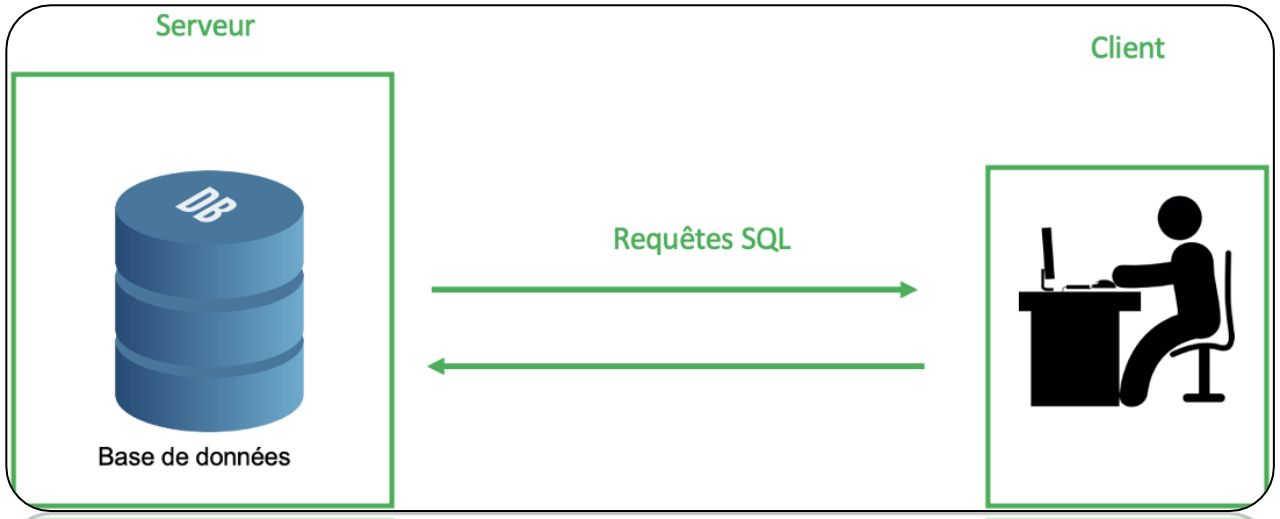
```
mysql> INSERT INTO etudiant (nom, email, note) VALUES
 ('Alami Samir', 'alamisa@etd.uae.ac.ma', 14);
Query OK, 1 row affected (0.05 sec)
```

```
mysql> select * from etudiant;
+----+-------------+-----------------------+------+
| id | nom         | email                 | note |
+----+-------------+-----------------------+------+
|  1 | Alami Samir | alamisa@etd.uae.ac.ma |   14 |
+----+-------------+-----------------------+------+
1 row in set (0.00 sec)
```

Manipulation de DB : SQL Requests

Structured Query Language (SQL) :

- Langage de définition de données (LDD/DDL)
  - création de relations : CREATE TABLE
  - modification de relations: ALTER TABLE
  - suppression de relations: DROP TABLE
  - vues, index …. : CREATE VIEW ...
- Langage de manipulation de données (LMD /DML)
  - insertion de tuples: INSERT
  - mise à jour des tuples: UPDATE
  - suppression de tuples: DELETE
- Langage de requêtes (LMD/DML)
  - SELECT ……. FROM ..... WHERE .....



Module « Algèbre relationnelle & Langage SQL » en IIDL-S4